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Technical Reference Series

SQL Server Replication

Top 200 On-Premises Troubleshooting Scenarios

Snapshot • Transactional • Merge • Peer-to-Peer

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About This Reference

This document catalogs 200 real-world SQL Server replication problems a DBA is likely to face on-premises across Snapshot, Transactional, Merge, and Peer-to-Peer topologies. It is organized by functional area and written for quick interview recall as well as production triage. Each scenario follows a consistent four-part structure:

- **Issue:** The symptom or error exactly as a DBA observes it.
- **Root Cause:** The underlying reason the issue occurs.
- **Analysis:** How to reason about and confirm the diagnosis.
- **Fix:** The concrete corrective action to resolve it.

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Section 1 Configuration & Setup

20 scenarios

001 sp_replicationdboption fails with 'Only members of the sysadmin fixed server role can perform this operation'.

Root Cause: The login enabling replication on the database lacks sysadmin membership, which is required to mark a database for publication.

Analysis: Enabling a database for replication modifies server-level metadata (syspublications, distribution links), so it cannot be delegated to db_owner alone. Verify with SELECT IS_SRVROLEMEMBER('sysadmin').

FIX Grant sysadmin temporarily or have a sysadmin run
EXEC sp_replicationdboption @dbname='MyDB', @optname='publish', @value='true'.

002 Configuring the Distributor fails with error 21633 'This SQL Server instance is not installed with replication components'.

Root Cause: Replication was not selected during SQL Server setup, so replication binaries (replrec.dll, sqlrepss.dll) and system tables are missing.

Analysis: Check SELECT SERVERPROPERTY('IsFullTextInstalled') is unrelated; instead confirm replication under SELECT * FROM sys.dm_server_registry or run setup /Action=RunDiscovery. The feature must be added.

FIX Run SQL Server setup, choose 'Add features', and install Replication.
No full reinstall is needed; a restart is not required for most versions.

003 Distributor configuration wizard cannot create the distribution database because the snapshot folder is inaccessible.

Root Cause: The default snapshot share (e.g. \\SERVER\ReplData) either does not exist or the SQL Server Agent service account lacks NTFS/share permissions.

Analysis: Snapshot and Distribution agents write BCP files to this folder. If it is a local path, remote Subscribers cannot pull. Test access from the agent account context.

FIX Create a UNC share, grant the Agent/agent-process account Read/Write on both share and NTFS, and set @working_directory accordingly in sp_adddistributor.

004 Publisher cannot use the remote Distributor: 'The remote server does not exist or has not been designated as a valid Publisher'.

Root Cause: The remote Distributor was configured but the Publisher was not registered on it, or the password in sp_adddistributor did not match.

Analysis: On the Distributor run sp_helpdistpublisher to confirm the Publisher entry. The @password used at the Publisher must equal what the Distributor expects.

FIX On the Distributor, EXEC sp_adddistpublisher with matching @login/@password; on the Publisher, point to the remote Distributor with the same admin password.

005 Creating a publication fails: 'The database is not enabled for publication'.

Root Cause: sp_addpublication was called before the database was marked as a publishing database.

Analysis: The 'publish' option must be true before any publication object can be created. Confirm with SELECT name, is_published FROM sys.databases.

FIX Run sp_replicationdboption @optname='publish', @value='true' (and 'merge publish' for merge) prior to sp_addpublication.

006 Transactional publication requires a Log Reader Agent but none is created.

Root Cause: The first transactional/P2P publication in a database automatically creates the Log Reader Agent job; if job creation fails (Agent account issue), the publication is left half-configured.

Analysis: Check msdb job history for the Log Reader job. A missing agent means no commands ever reach the distribution database.

FIX Run sp_addlogreader_agent with a valid @job_login/@job_password, or drop and recreate the publication once the SQL Agent proxy/account is valid.

007 Cannot add an article: 'Table does not have a primary key' for transactional replication.

Root Cause: Transactional replication requires a primary key on every published table to uniquely identify rows for command generation.

Analysis: Without a PK, the Log Reader cannot build UPDATE/DELETE statements keyed to a row. Merge uses rowguid instead but transactional is strict.

FIX Add a PRIMARY KEY to the table before sp_addarticle, or exclude the table. For heaps consider a unique clustered index promoted to PK.

008 Subscription creation fails with 'The subscription could not be created because the publication is not enabled for that subscription type'.

Root Cause: @allow_push / @allow_pull / @immediate_sync / @independent_agent flags on the publication do not permit the chosen subscription model.

Analysis: Push and pull have different agent placement. Also @allow_anonymous and @immediate_sync interact: anonymous subscriptions require immediate_sync=true.

FIX Reconfigure the publication with sp_changepublication to enable the required subscription type, or create the subscription matching the enabled type.

009 Snapshot folder path differs between Publisher and Subscriber, breaking pull subscriptions.

Root Cause: The default snapshot location is local to the Distributor; remote pull Subscribers need a UNC path they can reach.

Analysis: Push agents run on the Distributor and can use local paths, but pull agents run on the Subscriber and require an accessible network share.

> **FIX** Set @alt_snapshot_folder or configure the publication's snapshot to a UNC share reachable by all Subscribers via sp_changepublication @property='alt_snapshot_folder'.

010 Replication setup succeeds but the Snapshot Agent never starts on schedule.

Root Cause: The Snapshot Agent job schedule is disabled or set to run on demand, so no initial snapshot is generated.

Analysis: New publications may create the job with no schedule. Subscribers waiting for initialization will stay in 'waiting for snapshot' indefinitely.

> **FIX** Attach a schedule to the Snapshot Agent job or start it manually; confirm @sync_method and that the job owner is valid.

011 'Replication-Distribution' subsystem fails to load; agent jobs error immediately.

Root Cause: The SQL Server Agent proxy or subsystem DLLs are corrupt/missing, or the Agent service account changed without re-granting rights.

Analysis: Check msdb.dbo.syssubsystems and the Agent error log. Replication agents run as Agent subsystem steps; a broken subsystem stops all of them.

> **FIX** Reinstall replication components or run sp_verify_subsystems; ensure the Agent service account has 'Log on as a service' and folder rights.

012 Publication uses 'sync with backup' but log is still truncating too aggressively.

Root Cause: The sync-with-backup option was set on the distribution database but not on the publication database, or log backups are not running.

Analysis: sync with backup delays log truncation until the Log Reader has hardened commands into distribution and both are backed up. Misconfiguration risks data loss on failover.

> **FIX** Set sp_replicationdboption @optname='sync_with_backup' on the correct database and ensure regular log backups on both Publisher and Distributor.

013 Immediate updating / updatable subscription setup fails with MSMQ or linked server errors.

Root Cause: Queued/immediate updating subscriptions rely on triggers and linked servers (or Message Queuing) that were never configured.

Analysis: Updatable subscriptions are deprecated but still seen; they need a loopback linked server and MS DTC for immediate updating. Missing DTC breaks 2-phase commit.

> **FIX** Configure MS DTC and the required linked servers, or migrate off updatable subscriptions to a supported topology such as P2P or merge.

014 Cannot enable replication on a database in an Availability Group at the secondary.

Root Cause: Publisher redirection for AG was not configured, so the Distributor still points to the original Publisher instance name after failover.

Analysis: Replication is AG-aware only if you register the AG listener as the Publisher via sp_redirect_publisher. Otherwise agents fail after failover.

> **FIX** Run sp_validate_replica_hosts_as_publishers and sp_redirect_publisher to map the original Publisher to the AG listener.

015 Distribution database placed on a drive that later fills, halting all replication.

Root Cause: The distribution database and its log were created on an undersized volume; retention of undelivered commands grew MSrepl_commands.

Analysis: MSrepl_commands and MSrepl_transactions store every pending command until delivered and cleaned up. High latency or offline Subscribers cause explosive growth.

> **FIX** Move distribution DB files to a larger volume, lower @retention, fix the delivery bottleneck, and let the distribution cleanup job catch up.

016 Publication defined with @sync_method='native' fails on heterogeneous or case-sensitive Subscribers.

Root Cause: Native BCP format and collation assumptions differ; character mode is required for some cross-version or non-SQL targets.

Analysis: Native mode is faster but format-sensitive. Cross-collation or older Subscribers may reject native BCP files during snapshot apply.

> **FIX** Set @sync_method='character' (or 'concurrent_c') for compatibility, regenerate the snapshot, and reinitialize subscriptions.

017 Filtered publication (row/column filter) omits columns a Subscriber query needs, and adding them later forces reinit.

Root Cause: Vertical/horizontal filters were defined too narrowly at publication creation without accounting for downstream requirements.

Analysis: Changing article filters is a schema-affecting change that generally invalidates the existing snapshot and requires reinitialization.

> **FIX** Plan filters carefully; to change, use sp_articlecolumn / sp_articlefilter, regenerate the snapshot, and reinitialize affected subscriptions.

018 Two publications on the same table cause 'article already exists in another publication with conflicting properties'.

Root Cause: The same base table is published in multiple publications with incompatible article settings (e.g. different schema options).

Analysis: Articles can be shared across publications only if their key properties match. Divergent @schema_option or filter settings collide.

> **FIX** Align article properties across publications, or use distinct article names/destination objects; consolidate where possible.

019 Replication configured but sp_removedbreplication is needed after a failed teardown leaves orphaned objects.

Root Cause: A partial drop left replication system objects, triggers, or metadata behind, blocking a clean re-setup.

Analysis: Orphaned MSpub/MSsub objects or replication triggers prevent re-enabling. sp_removedbreplication forcibly clears replication metadata from a database.

> **FIX** Run EXEC sp_removedbreplication 'MyDB' on the affected database, then reconfigure replication from scratch.

020 Wrong Distributor admin password recorded, so Publisher-Distributor handshake silently fails.

Root Cause: The distributor_admin (repl_distributor) password stored at the Publisher does not match the Distributor.

Analysis: The linked server 'repl_distributor' uses this credential. A mismatch produces authentication errors in agent logs without an obvious cause.

> **FIX** Reset with sp_changedistributor_password on the Distributor and update the Publisher via sp_adddistributor / sp_changedistributor_password consistently.

Section 2 Snapshot Agent

15 scenarios

021 Snapshot Agent fails: 'Cannot bulk load. The file does not exist or you don't have file access rights'.

Root Cause: The Agent process account cannot read/write the snapshot folder, or the path is local while the agent runs remotely.

Analysis: The Snapshot Agent BCPs schema and data into .sch/.bcp/.idx files. Any permission or path issue aborts the whole snapshot.

> **FIX** Grant the Snapshot Agent account full control on the snapshot share/NTFS and use a UNC path reachable by the agent.

022 Snapshot Agent hangs at 'Bulk copying data into table' for a very large table.

Root Cause: A huge table with no batching, plus locking/blocking, stalls the BCP-out phase.

Analysis: By default the Snapshot Agent takes shared locks during snapshot generation (concurrent snapshot for transactional reduces this). Long BCP-out on billion-row tables looks like a hang.

> **FIX** Use concurrent snapshot processing, schedule during low activity, pre-size tempdb, and consider initializing from backup for very large publications.

023 Snapshot generation blocks user activity with long-held Sch-S/Shared locks.

Root Cause: Non-concurrent snapshot method holds locks on published tables for the duration of BCP-out.

Analysis: For transactional replication, concurrent snapshots hold locks only briefly. Merge and some options still take longer locks. Blocking chains form on OLTP tables.

> **FIX** Ensure @sync_method='concurrent' for transactional publications and run snapshots off-peak; monitor with sys.dm_tran_locks.

024 Snapshot Agent error: 'The process could not create file ... because the directory is full or the disk is full'.

Root Cause: The snapshot volume ran out of space while writing BCP data files.

Analysis: Snapshots of large publications can temporarily need space equal to the published data size (or more with indexes/format files).

> **FIX** Free space or point @alt_snapshot_folder to a larger volume; compress snapshot files (@compress_snapshot) if network transfer is the concern.

025 Compressed snapshot (.cab) fails to apply on the Subscriber.

Root Cause: @compress_snapshot=true was set but the alternate snapshot folder or FTP/UNC delivery is misconfigured, or the CAB exceeds size limits.

Analysis: CAB compression has a 2GB per-cab historical limit and requires the alt snapshot folder. Distribution Agent must be able to locate and decompress it.

> **FIX** Verify the alt snapshot folder path, keep CAB sizes within limits, and confirm the Distribution/Merge Agent can access the compressed files.

026 Snapshot Agent regenerates a full snapshot every run, flooding the network.

Root Cause: @immediate_sync=true keeps the snapshot always available and may regenerate on each schedule; or a script drops/recreates the publication.

Analysis: With immediate_sync, the snapshot is kept current so new anonymous Subscribers can initialize anytime, which can mean frequent regeneration.

> **FIX** Set @immediate_sync=false when you don't need always-available snapshots, and schedule the Snapshot Agent only when reinitialization is required.

027 Snapshot applies schema but 'permission denied' when creating objects on the Subscriber.

Root Cause: The Distribution/Merge Agent's Subscriber connection account lacks rights to create tables/indexes in the subscription database.

Analysis: Applying a snapshot executes .sch scripts (CREATE TABLE, indexes, constraints). Insufficient rights fail mid-apply, leaving partial schema.

> **FIX** Grant the Subscriber connection account db_owner on the subscription database (or the minimal required DDL rights) and reinitialize.

028 Snapshot Agent fails to script an object that uses an unsupported feature for the Subscriber version.

Root Cause: The article uses a datatype or feature (e.g. newer column store, computed column semantics) not supported on a down-level Subscriber.

Analysis: Replication supports N-2 version interop with rules; scripting features the Subscriber can't parse fails snapshot apply.

> **FIX** Adjust @schema_option to omit unsupported clauses, or upgrade the Subscriber; test the generated .sch script on the target version.

029 Snapshot files left behind consume disk after successful initialization.

Root Cause: The distribution cleanup / snapshot cleanup does not remove alternate-folder snapshots, or @immediate_sync keeps them.

Analysis: Standard cleanup handles the default folder based on retention; alt snapshot folders and manually delivered snapshots may not be auto-cleaned.

> **FIX** Implement housekeeping to purge stale snapshot folders, lower retention, and disable immediate_sync where always-available snapshots aren't needed.

030 Snapshot Agent reports success but Subscriber still shows 'no snapshot available'.

Root Cause: The Distribution Agent is looking in a different snapshot folder than where the Snapshot Agent wrote (default vs alternate).

Analysis: A mismatch between @snapshot_folder used at generation and the path the Distribution Agent reads causes 'snapshot not available' despite a good run.

> **FIX** Align the alternate snapshot folder settings on publication and subscription, then rerun the Snapshot and Distribution Agents.

031 Concurrent snapshot fails: 'Another snapshot for this publication is already running'.

Root Cause: A previous Snapshot Agent instance did not exit cleanly and left a running job or lock.

Analysis: Only one snapshot per publication runs at a time. A hung agent or duplicate schedule causes the conflict.

> **FIX** Stop the stuck Snapshot Agent job, clear its history, verify no orphaned session, then restart the snapshot.

032 Snapshot includes pre/post scripts that fail on the Subscriber.

Root Cause: @pre_snapshot_script / @post_snapshot_script reference objects or paths not valid on the Subscriber.

Analysis: These scripts run during snapshot apply; a failure aborts initialization and marks the subscription failed.

> **FIX** Validate the pre/post scripts against the Subscriber, correct paths/objects, and reinitialize.

033 Initializing from backup is desired but Subscriber keeps waiting for a snapshot.

Root Cause: @allow_initialize_from_backup was not enabled on the publication, so the Subscriber can't use a restored backup as its baseline.

Analysis: Init-from-backup avoids huge snapshot transfers for large databases, but must be explicitly allowed and the subscription created with @sync_type='initialize with backup'.

FIX Set sp_changepublication @property='allow_initialize_from_backup', @value='true', restore a recent backup on the Subscriber, and add the subscription with the backup sync type.

034 Snapshot Agent very slow due to many small articles scripted serially.

Root Cause: Hundreds of articles each require schema scripting and BCP, and the agent processes them with limited parallelism.

Analysis: Overhead per article (open/close BCP, format files) dominates when article count is high, even if data volume is small.

FIX Split into multiple publications to parallelize, or reduce article count; use init-from-backup for large baselines.

035 Snapshot Agent uses too much tempdb and fails with 'Could not allocate space for object in tempdb'.

Root Cause: Concurrent snapshot and sort operations for index creation spill heavily into tempdb.

Analysis: Rebuilding clustered/nonclustered indexes during snapshot apply and BCP sorting can exhaust an undersized tempdb.

FIX Pre-size tempdb with multiple files, add space, and schedule snapshots when tempdb pressure is low.

Section 3 Log Reader Agent

11 scenarios

036 Log Reader Agent stops with 'The process could not execute sp_replcmds on Publisher'.

Root Cause: The Log Reader account lost access to the published database, or the transaction log is inaccessible/corrupt for the marked LSN.

Analysis: sp_replcmds scans the log for replicated transactions. Permission loss, log corruption, or a database in an unusual state (recovery pending) halts it.

FIX Verify the Log Reader account has access and db is online; check DBCC LOGINFO/errorlog; if corruption, restore or run sp_replrestart carefully.

037 Transaction log on the Publisher grows uncontrollably despite frequent log backups.

Root Cause: The Log Reader Agent is not harvesting replicated transactions, so the log records stay marked as pending replication and cannot be truncated.

Analysis: log_reuse_wait_desc shows REPLICATION. Until the Log Reader copies commands into distribution, log truncation is blocked even in FULL/SIMPLE.

> **FIX** Start/repair the Log Reader Agent; if replication is broken and won't recover, use sp_repldone/sp_replflush cautiously to release the log, then reinitialize.

038 log_reuse_wait_desc = REPLICATION even though the database is not published anymore.

Root Cause: Replication was removed improperly, leaving the database still flagged as replicated with pending log records.

Analysis: Orphaned replication state pins the log. This is common after dropping publications without sp_removedbreplication.

> **FIX** Run EXEC sp_removedbreplication; if it persists, EXEC sp_repldone @xactid=NULL, @xact_seqno=NULL, @numtrans=0, @time=0, @reset=1 then back up the log.

039 Log Reader latency spikes; commands accumulate in the log before reaching distribution.

Root Cause: High transaction volume, large batches, or an undersized -ReadBatchSize/-PollingInterval slow the reader.

Analysis: The Log Reader reads in batches; huge single transactions or index maintenance floods it. Latency shows in Replication Monitor as Publisher-to-Distributor.

> **FIX** Tune -ReadBatchSize, -MaxCmdsInTran, and -PollingInterval; avoid giant transactions; batch index maintenance to reduce replicated volume.

040 Log Reader Agent errors: 'The row was not found at the Subscriber when applying the replicated command' originates from a large log batch.

Root Cause: Actually a Distribution Agent error, but triggered by commands the Log Reader forwarded for rows already changed out-of-band at the Subscriber.

Analysis: Someone modified Subscriber data directly; the replicated UPDATE/DELETE can't find the row. Distinguish Log Reader vs Distribution errors by agent name.

> **FIX** Stop direct Subscriber writes; use -SkipErrors 20598 sparingly or reinitialize the affected article to resync.

041 Only one Log Reader per published database becomes a bottleneck for many publications.

Root Cause: By design there is a single Log Reader Agent per publishing database regardless of how many publications exist.

Analysis: All publications on a database share one log scan. A hot database with many articles can saturate the single reader.

FIX Reduce replicated write volume, split hot tables into separate databases if architecture allows, and tune reader batch parameters.

042 Log Reader Agent fails after a minimally-logged bulk operation on a published table.

Root Cause: Bulk operations under SIMPLE/BULK_LOGGED can leave the Log Reader unable to reconstruct commands if not fully logged for replication.

Analysis: Published tables force full logging for replicated operations, but misconfiguration or trace flags can interfere, breaking command generation.

FIX Ensure published databases use appropriate recovery; avoid unsupported bulk shortcuts on published tables; reinitialize if the log stream is broken.

043 Log Reader stuck reprocessing the same transaction after an agent crash.

Root Cause: The last delivered LSN was not committed to MSlogreader_history, so on restart the agent re-scans from an earlier point.

Analysis: Idempotent delivery normally prevents duplicates, but a crash between distribution insert and history update causes reprocessing and apparent stalls.

FIX Let it complete the re-scan; if it loops, check distribution DB health and consider validating with sp_replmonitorhelppublication.

044 Error 18752: 'Only one Log Reader Agent or log-related procedure can connect to a database at a time'.

Root Cause: A second process (another Log Reader, sp_replcmds, or a stuck session) is already reading the log.

Analysis: The log-scan interface is single-writer. Duplicate agents or leftover sessions collide with 18752.

FIX Kill the orphaned session, ensure only one Log Reader job per database, and restart the agent.

045 Log Reader Agent cannot start after Publisher restore/attach: 'The publication does not exist' or LSN mismatch.

Root Cause: Restoring the published database over an existing one desynchronizes replication metadata and the log chain.

Analysis: Replication metadata lives in the user database and msdb/distribution; a restore that doesn't preserve KEEP_REPLICATION breaks the linkage.

FIX Restore WITH KEEP_REPLICATION when supported, or remove and reconfigure replication and reinitialize subscriptions.

046 Log Reader Agent -Continuous mode silently exits after a severe error and doesn't restart.

Root Cause: A non-retryable error terminated the agent, and no job restart step or alerting caught it.

Analysis: Continuous agents are meant to run indefinitely, but fatal errors stop the job; without monitoring, latency grows unnoticed.

> **FIX** Add job restart-on-failure steps, configure Replication Monitor thresholds/alerts, and diagnose the fatal error from the agent output file.

Section 4 Distribution Agent

20 scenarios

047 Distribution Agent fails: 'The row was not found at the Subscriber when applying the replicated command' (Error 20598).

Root Cause: The target row was modified or deleted directly on the Subscriber, so the replicated UPDATE/DELETE cannot locate it.

Analysis: Transactional Subscribers should be read-only. Out-of-band changes create data divergence that surfaces as 20598 at delivery.

> **FIX** Fix Subscriber write permissions; to recover, reinitialize the article, or as a last resort use the -SkipErrors 20598 profile and then validate/repair.

048 Distribution Agent fails: 'Violation of PRIMARY KEY constraint ... Cannot insert duplicate key' (Error 2627).

Root Cause: A replicated INSERT targets a key that already exists on the Subscriber, often from prior partial delivery or manual inserts.

Analysis: Duplicate-key on INSERT signals divergence or reapplied commands. Common after skipping errors previously or reseeding identities incorrectly.

> **FIX** Reconcile the row, reinitialize the article, or use sp_browsereplcmds to inspect the offending command; avoid blanket -SkipErrors.

049 Distribution Agent latency high; commands queue in MSrepl_commands for hours.

Root Cause: Slow Subscriber apply due to missing indexes, blocking, small -SubscriptionStreams, or network latency.

Analysis: Delivery rate lags generation rate. Replication Monitor shows Distributor-to-Subscriber latency climbing and undistributed command counts rising.

FIX Increase -SubscriptionStreams, add supporting indexes on the Subscriber, resolve blocking, and tune -CommitBatchSize/-CommitBatchThreshold.

050 Distribution Agent applies commands very slowly because it calls per-row stored procedures.

Root Cause: Article was created with call-format procedures but the Subscriber CPU/IO can't keep up with per-command RPCs.

Analysis: Transactional replication uses sp_MSins/upd/del procedures by default. On high-volume articles these RPCs become the bottleneck.

FIX Consider @schema_option to use XCALL/streaming, batch with -SubscriptionStreams, and ensure Subscriber has proper indexing and hardware.

051 Distribution Agent error: 'Could not find stored procedure sp_MSins_<table>' on the Subscriber.

Root Cause: The custom insert/update/delete procedures were dropped on the Subscriber or never created due to a failed snapshot.

Analysis: Transactional replication delivers via generated sp_MSins/upd/del procs. If they are missing, delivery fails immediately.

FIX Reinitialize the subscription to regenerate the procs, or manually recreate them; ensure no process is dropping replication objects.

052 Distribution Agent fails on a computed or timestamp column mismatch.

Root Cause: Schema drift: the Subscriber table definition differs from what the replicated command expects (extra/missing/typed columns).

Analysis: Manual ALTERs on the Subscriber, or non-replicated DDL, cause the delivered command's column list to mismatch the target.

FIX Realign the Subscriber schema to the published article, or reinitialize; enforce that schema changes flow only through replication.

053 Push Distribution Agent competes for Distributor resources with many subscriptions.

Root Cause: All push agents run on the Distributor; dozens of subscriptions overwhelm CPU/memory there.

Analysis: Push centralizes agent execution on the Distributor. Large fan-out topologies can saturate it.

FIX Convert high-count topologies to pull subscriptions (agents run on Subscribers), or scale out the Distributor and stagger schedules.

054 Distribution Agent connection to Subscriber times out intermittently over WAN.

Root Cause: Network instability or an aggressive -LoginTimeout/-QueryTimeout causes drops during delivery over a slow link.

Analysis: WAN Subscribers experience packet loss/latency; default timeouts may be too tight, causing repeated agent restarts.

> **FIX** Increase -QueryTimeout and -LoginTimeout, enable continuous mode with retries, and consider compression or a closer Distributor.

055 Distribution Agent delivers out of order after -SubscriptionStreams was increased.

Root Cause: Multiple streams parallelize delivery but constraint/FK dependencies across streams cause transient ordering violations.

Analysis: SubscriptionStreams boosts throughput but requires that inter-transaction dependencies tolerate parallelism; heavy FK chains can fail.

> **FIX** Reduce streams for FK-heavy articles, or ensure delivery ordering via appropriate article grouping; test throughput vs correctness.

056 Distribution Agent fails applying a large text/image/varchar(max) update ('String or binary data would be truncated').

Root Cause: Subscriber column length is smaller than the Publisher's, often due to a manual schema change or a mismatched initial script.

Analysis: Truncation (Error 8152/2628) means the target column can't hold the replicated value. Divergent DDL is the usual cause.

> **FIX** Align column definitions, widen the Subscriber column, and reinitialize the affected article.

057 Distribution Agent stuck 'Delivering replicated transactions' with zero progress and no error.

Root Cause: It is blocked on the Subscriber by a long-running query or lock, or waiting on a large single transaction to commit.

Analysis: A single massive replicated transaction (e.g. one big DELETE) must apply atomically; the agent appears frozen until it finishes or is blocked.

> **FIX** Investigate Subscriber blocking (sys.dm_exec_requests), break large publisher transactions into batches, and tune commit batch settings.

058 Distribution Agent history bloats msdb/distribution and slows Replication Monitor.

Root Cause: Verbose agent -HistoryVerboseLevel and long history retention accumulate millions of rows in MSdistribution_history.

Analysis: High verbosity is useful for debugging but hurts performance and Monitor responsiveness at scale.

> **FIX** Lower -HistoryVerboseLevel to 1, ensure the history cleanup job runs, and keep distribution history retention reasonable.

059 Distribution Agent fails: 'Cannot execute as the database principal because the principal does not exist' on the Subscriber.

Root Cause: The agent connects as a login mapped to a missing/orphaned user, or EXECUTE AS context is broken on delivery procs.

Analysis: Orphaned users after a Subscriber restore, or dropped mappings, break the security context needed to run sp_MSins/upd/del.

> **FIX** Fix orphaned users (ALTER USER ... WITH LOGIN), ensure the agent login maps correctly, and grant execute on the delivery procs.

060 Distribution Agent applies a delete that cascades unexpectedly on the Subscriber.

Root Cause: FK relationships with ON DELETE CASCADE exist on the Subscriber but not accounted for in publication design.

Analysis: Replicated deletes trigger Subscriber-side cascades, removing rows replication expects to manage, causing later 20598 errors.

> **FIX** Replicate constraints consistently via @schema_option, avoid extra cascading FKs on the Subscriber, and reconcile affected rows.

061 Multiple Subscribers share one Distribution Agent (independent_agent=false) and one failure blocks all.

Root Cause: @independent_agent=false creates a shared agent per publication; a single Subscriber problem stalls delivery to all.

Analysis: Shared agents were historically default; they serialize delivery and reduce isolation between Subscribers.

> **FIX** Use independent agents (@independent_agent=true) so each subscription has its own agent and failures are isolated.

062 Distribution Agent -SkipErrors was set globally and silently dropped legitimate data.

Root Cause: An operator added -SkipErrors to the agent profile to clear a stuck agent, masking real divergence.

Analysis: SkipErrors bypasses specified error numbers, moving past bad commands without applying them, which compounds data drift over time.

> **FIX** Remove blanket SkipErrors, validate data with tablediff/sp_publication_validation, and reinitialize to restore consistency.

063 Distribution Agent fails delivering to a Subscriber whose collation differs from the Publisher.

Root Cause: Collation mismatch causes comparison/insert errors for string keys and predicates in delivery procs.

Analysis: Cross-collation replication is allowed but predicate evaluation in sp_MSupd/del can misbehave, and unique key comparisons may differ.

> **FIX** Standardize collation, or explicitly handle it in article design; test key comparisons; reinitialize after alignment.

064 Distribution Agent shows 'skipped' transactions after a Distributor cleanup ran mid-delivery.

Root Cause: The distribution cleanup job removed commands still needed because retention was too short relative to Subscriber downtime.

Analysis: If a Subscriber is offline longer than the max distribution retention, its pending commands are purged and it can no longer catch up.

> **FIX** Increase distribution @retention/@max_distretention, reinitialize the lagging Subscriber, and monitor Subscriber uptime.

065 Distribution Agent error 'The publication ... does not exist' after publication was dropped and recreated.

Root Cause: The subscription still references the old publication identity/metadata that no longer matches.

Analysis: Dropping and recreating a publication invalidates existing subscriptions; agents can't map to the new publication_id.

> **FIX** Drop and recreate the subscriptions against the new publication, generate a fresh snapshot, and reinitialize.

066 Distribution Agent applies commands successfully but the Subscriber shows a deadlock victim rollback each cycle.

Root Cause: The agent is repeatedly chosen as the deadlock victim by concurrent Subscriber workload, rolling back and retrying.

Analysis: Error 1205 on the delivery session forces a rollback; the agent retries, wasting cycles and adding latency.

> **FIX** Reduce competing Subscriber activity during delivery, add indexes to shorten lock duration, and consider raising the agent's deadlock priority.

Section 5 Merge Agent

20 scenarios

067 Merge Agent fails: 'The schema script could not be propagated to the subscriber'.

Root Cause: A DDL change or article schema script failed to apply on the Subscriber, often due to permissions or an incompatible object.

Analysis: Merge propagates schema changes as generation batches; a single failed script blocks the merge session.

> **FIX** Inspect MSmerge_history for the failing script, correct the object/permission on the Subscriber, and re-run the Merge Agent.

068 Merge replication generates excessive metadata; MSmerge_contents/MSmerge_tombstone grow huge.

Root Cause: Retention is high and merge tracking retains change/tombstone metadata for every row change beyond what is needed.

Analysis: Merge tracks lineage and tombstones for conflict detection. Long retention plus high churn bloats metadata and slows merges.

> **FIX** Lower publication @retention (subject to your longest offline Subscriber), ensure the merge metadata cleanup runs, and rebuild indexes on merge system tables.

069 Merge Agent conflicts resolve unexpectedly, overwriting the 'wrong' side.

Root Cause: The configured conflict resolver (default: Publisher wins) does not match business intent, or article priorities are misconfigured.

Analysis: Merge uses column/row-level tracking and a resolver. Default resolvers favor the Publisher or higher-priority Subscriber; surprises stem from resolver choice.

FIX Choose an appropriate business/COM resolver, set article-level conflict tracking (column vs row), and review conflicts in the conflict viewer.

070 Merge Agent very slow enumerating changes for a Subscriber that was offline a long time.

Root Cause: A large backlog of generations must be enumerated and downloaded, and the metadata scan is expensive.

Analysis: Merge enumerates changes by generation. Long offline periods create many generations to process, causing long first-sync times.

FIX Reinitialize very-stale Subscribers instead of catching up, tune -DownloadGenerationsPerBatch/-UploadGenerationsPerBatch, and keep retention aligned to offline windows.

071 Merge replication rowguid column caused application inserts to fail.

Root Cause: Merge added a rowguid column with a default; application inserts that specify column lists or expect a fixed shape break.

Analysis: Merge requires a uniqueidentifier ROWGUIDCOL. If the app does INSERT without column lists or has SELECT * dependencies, the extra column surprises it.

FIX Add an explicit rowguid column beforehand, use column lists in inserts, and educate the app team; the column is mandatory for merge.

072 Merge Agent fails on identity ranges: 'The insert failed. It conflicted with an identity range check constraint'.

Root Cause: The Subscriber exhausted its assigned identity range and automatic range management did not allocate a new one.

Analysis: Merge assigns identity ranges per node via check constraints. If reassignment fails (permissions, threshold), inserts hit the range ceiling.

FIX Ensure @identity_range_management='auto', grant the Merge Agent rights to adjust ranges, and use sp_adjustpublisheridentityrange if needed.

073 Merge filters (parameterized/join filters) cause slow enumeration and 'non-overlapping partition' errors.

Root Cause: Complex join/parameterized filters with HOST_NAME()/SUSER_SNAME() force heavy partition evaluation per Subscriber.

Analysis: Dynamic filters partition data per Subscriber; deep join filter chains are expensive and can misclassify rows if partitions overlap incorrectly.

FIX Simplify filter joins, set @partition_options appropriately (non-overlapping/single), pre-compute partitions, and index filter columns.

074 Merge Agent error: 'The Merge Agent failed because the schema of the article at the Publisher and Subscriber do not match'.

Root Cause: A schema change was applied at one node without flowing through merge DDL propagation.

Analysis: Merge tolerates schema changes only when made via supported ALTER paths that propagate. Out-of-band DDL desynchronizes the article shape.

FIX Reapply changes through the Publisher so they propagate, or reinitialize; avoid direct DDL on merge articles.

075 Web synchronization for merge fails with HTTP 500 / SSL errors.

Root Cause: The IIS Replication Listener (replisapi.dll), certificate, or SQL Server CE agent configuration is broken.

Analysis: Web sync tunnels merge over HTTPS via replisapi. Certificate binding, app-pool identity, or ISAPI config problems cause 500-series errors.

FIX Verify the IIS virtual directory, SSL certificate, app-pool identity permissions, and rerun the Web Synchronization Wizard.

076 Merge Agent uploads changes but downloads nothing; one-way behavior unexpected.

Root Cause: The article was set to @subscriber_upload_options that restrict download, or the subscription is download-only/upload-only by design.

Analysis: Merge articles can be bidirectional, download-only, or upload-restricted. Misconfigured upload options create surprising one-directional sync.

FIX Review sp_addmergearticle @subscriber_upload_options and subscription type; set the intended directionality and reinitialize if needed.

077 Merge triggers (ins/upd/del) missing on a table, so changes aren't tracked.

Root Cause: The merge tracking triggers were dropped or disabled, or a table was added without proper merge article setup.

Analysis: Merge relies on triggers to populate MSmerge_contents. Missing triggers mean silent data loss between nodes.

FIX Reinitialize the article to recreate triggers, and ensure no maintenance disables/drops replication triggers.

078 Merge Agent deadlocks with application on merge system tables during sync.

Root Cause: Concurrent user DML and the Merge Agent contend on MSmerge_contents and base tables.

Analysis: High concurrency plus large generation batches increase lock contention, producing deadlocks (Error 1205) during merges.

> **FIX** Schedule merges off-peak, reduce generation batch sizes, add appropriate indexes, and consider retry logic on the agent.

079 Merge replication logical records break referential integrity across related tables.

Root Cause: Related rows in a logical record were not grouped, so parent/child arrive in different generations.

Analysis: Without logical records, merge may apply a child before its parent, violating FKs during sync.

> **FIX** Define logical records (sp_addmergefilter with @filter_type) to keep related rows together, or relax constraint timing.

080 Merge Agent error 'The number of failed attempts has reached the maximum retry count'.

Root Cause: A persistent conflict/constraint failure recurs each session and exhausts the retry count.

Analysis: Merge retries transient failures; a deterministic error (bad resolver, constraint) never clears and trips the max-retry limit.

> **FIX** Diagnose the underlying constraint/conflict from MSmerge_history, fix the data or resolver, then re-run.

081 Precomputed partitions become stale, sending wrong rows to Subscribers.

Root Cause: @use_partition_groups is enabled but partition metadata wasn't refreshed after membership changes.

Analysis: Precomputed partitions speed enumeration but must be maintained; membership changes (e.g. a user moves regions) can misroute data.

> **FIX** Run sp_mergemetadataretentioncleanup and recompute partitions, or disable precomputed partitions for volatile filters.

082 Merge subscription expires: 'The subscription(s) have expired or do not exist'.

Root Cause: The Subscriber did not synchronize within the publication retention period, so its metadata was cleaned up.

Analysis: Retention defines the max time between syncs. Beyond it, the Subscriber is considered expired and must reinitialize.

FIX Increase retention to cover real offline windows, monitor last-sync times, and reinitialize expired Subscribers.

083 Merge Agent fails converting a datatype not supported for merge (e.g. certain LOB or spatial edge cases).

Root Cause: An article contains a datatype/feature with merge limitations, breaking metadata tracking or apply.

Analysis: Merge has more datatype/feature restrictions than transactional (e.g. some computed columns, certain constraints).

FIX Exclude or transform unsupported columns, or choose transactional replication for those tables.

084 Global vs local subscription priorities cause unexpected conflict winners in a multi-Subscriber merge.

Root Cause: Subscription priority (server subscriptions vs client) determines conflict resolution, and it was left at defaults.

Analysis: Server (global) subscriptions carry priority values; client subscriptions use Publisher-wins. Mixed topologies surprise operators.

FIX Assign explicit priorities to server subscriptions to reflect business precedence and document expected conflict outcomes.

085 Merge metadata cleanup job removes tombstones needed by a slow Subscriber, causing missed deletes.

Root Cause: Cleanup ran within retention but a Subscriber synced after tombstones were purged, so deletes weren't propagated.

Analysis: If a delete's tombstone is cleaned before a lagging Subscriber syncs, that Subscriber never learns of the delete.

FIX Align retention to the slowest Subscriber's sync interval; reinitialize Subscribers that exceed retention.

086 Merge Agent connectivity via SQL Server Compact edition clients fails after platform changes.

Root Cause: SQL CE / mobile client sync stack deprecated or mismatched with the current SQL Server merge features.

Analysis: Legacy mobile merge scenarios depend on specific client libraries; platform upgrades break the sync provider.

FIX Migrate off deprecated SQL CE merge to a supported sync approach (e.g. modern sync framework) or a supported client version.

Section 6 Distributor & Distribution DB

20 scenarios

087 Distribution database log grows very large and won't shrink.

Root Cause: Undelivered commands and open transactions (offline Subscribers, delivery lag) keep the distribution log active.

Analysis: MSrepl_transactions/MSrepl_commands retention plus delivery lag hold log space. Until cleanup and delivery advance, the log can't truncate.

FIX Resolve delivery lag, reduce retention, ensure the distribution cleanup and log backup (if FULL) run, then shrink once space frees.

088 'Agent history clean up: distribution' and 'Distribution clean up' jobs disabled, causing bloat.

Root Cause: Cleanup jobs were disabled or failing, so history and delivered commands accumulate indefinitely.

Analysis: The distribution cleanup job (sp_MSdistribution_cleanup) purges delivered commands past retention. Disabled cleanup bloats the distribution DB.

FIX Re-enable and verify the cleanup jobs, fix any errors, and let them purge; monitor MSrepl_commands row counts.

089 Distribution database corruption reported by DBCC CHECKDB.

Root Cause: IO subsystem errors or an unexpected shutdown corrupted distribution system tables.

Analysis: Distribution DB is a normal database and can corrupt. Corruption in MSrepl_commands halts all delivery.

FIX Restore the distribution DB from backup if available, or the safest recovery is to remove and reconfigure replication and reinitialize.

090 Retention set too high, so undelivered commands pile up and inflate the distribution database.

Root Cause: @max_distretention (default 72h) and @min_distretention keep delivered commands too long for the workload.

Analysis: Longer retention aids offline Subscribers but costs storage and slows cleanup/Monitor. High-volume systems feel this acutely.

> **FIX** Tune sp_changedistributiondb retention to balance offline tolerance against storage; typical is 24-72h.

091 Replication Monitor extremely slow to open with many publications/Subscribers.

Root Cause: Large MSdistribution_history and verbose agent logging make Monitor queries expensive.

Analysis: Monitor aggregates history across all agents; millions of history rows and high verbosity degrade its responsiveness.

> **FIX** Lower -HistoryVerboseLevel, ensure history cleanup runs, and index/maintain distribution history tables.

092 Remote Distributor becomes a single point of failure for multiple Publishers.

Root Cause: Centralizing distribution improves Publisher performance but concentrates risk on one Distributor.

Analysis: If the remote Distributor is down, all Publishers' Log Readers can't offload commands and Publisher logs grow.

> **FIX** Make the Distributor highly available (FCI/AG-backed storage), monitor it closely, and plan capacity for combined load.

093 Distribution database on tempdb-like fast disk fills during a reinitialization storm.

Root Cause: Mass reinitialization regenerates snapshots and floods distribution with commands simultaneously.

Analysis: Reinitializing many subscriptions at once spikes distribution IO and space usage far above steady state.

> **FIX** Stagger reinitializations, pre-size the distribution DB, and temporarily raise cleanup frequency.

094 Error 21879 / 'Could not connect to Distributor' after Distributor SQL restart.

Root Cause: The repl_distributor linked server or the distributor_admin credential was invalidated by a service restart or password rotation.

Analysis: Publisher-Distributor communication uses a stored credential; rotation without updating both sides breaks connectivity.

> **FIX** Reset with sp_changedistributor_password and confirm the repl_distributor linked server on the Publisher works.

095 Distribution cleanup deadlocks with Distribution Agents under heavy load.

Root Cause: The cleanup job and delivery agents contend on MSrepl_commands, producing deadlocks.

Analysis: Cleanup deletes delivered commands while agents read/insert; without good indexing this contention causes 1205 errors.

> **FIX** Schedule cleanup during lower activity, keep distribution system table indexes healthy, and consider smaller cleanup batches.

096 Multiple distribution databases needed but only one exists, causing contention.

Root Cause: All publications route through a single distribution database, serializing cleanup and history.

Analysis: A busy environment benefits from multiple distribution databases to spread IO and isolate publications.

> **FIX** Create additional distribution databases (sp_adddistributiondb) and assign publications across them to balance load.

097 Distributor properties show wrong SQL Agent account after a service account change.

Root Cause: The SQL Server Agent service account was changed at the OS level without updating replication agent security.

Analysis: Replication agents inherit or reference the Agent account; a mismatch causes folder/permission failures across agents.

> **FIX** Update the Agent account's rights (snapshot folder, distribution DB) and any agent job security/proxy settings.

098 Distribution database autogrowth set too small, causing frequent growth stalls under load.

Root Cause: Default small autogrowth increments trigger many growth events during command surges.

Analysis: Percentage or tiny fixed autogrowth on a high-write distribution DB causes IO stalls and fragmentation.

> **FIX** Set sensible fixed autogrowth (e.g. 512MB-1GB), pre-grow to expected size, and enable instant file initialization.

099 Cannot drop the Distributor: 'The Distributor has ... dependent publications/subscriptions'.

Root Cause: Publications and subscriptions still reference the Distributor, so it can't be removed.

Analysis: Distributor teardown requires removing all dependent replication objects first; leftovers block sp_dropdistributor.

FIX Drop subscriptions and publications, then sp_dropdistributiondb and sp_dropdistributor (optionally @no_checks=1 with care).

100 Distribution DB restored to a new server keeps pointing to the old Publisher name.

Root Cause: Server rename or migration wasn't reflected in replication metadata (MSdistpublishers, sysservers).

Analysis: Replication stores server names; moving the Distributor requires updating name references or agents fail to connect.

FIX Use sp_dropserver/sp_addserver to fix @@SERVERNAME, update MSdistpublishers references, or reconfigure replication cleanly.

101 MSrepl_errors table grows and Replication Monitor shows stale errors.

Root Cause: Error history isn't cleaned, and resolved errors persist, cluttering monitoring.

Analysis: MSrepl_errors stores agent failures; without cleanup it grows and old entries confuse triage.

FIX Ensure history/error cleanup jobs run; query MSrepl_errors with sp_browsereplcmds to correlate current issues only.

102 Distributor and Publisher on different SQL versions cause agent incompatibility.

Root Cause: The Distributor must be at a version equal to or higher than the Publisher; a lower Distributor version breaks support rules.

Analysis: Replication requires Distributor >= Publisher version. Violating this leads to unsupported/failing agent behavior.

FIX Upgrade the Distributor to meet/exceed the Publisher version per Microsoft's version interop matrix.

103 Distribution database in FULL recovery fills the log because no log backups run.

Root Cause: The distribution DB defaults can leave it in a recovery model where the log isn't backed up, so it grows.

Analysis: Distribution DB is typically SIMPLE, but if switched to FULL without log backups, the log balloons.

FIX Keep the distribution DB in SIMPLE recovery unless there's a specific reason; if FULL, schedule log backups.

104 sp_replmonitorhelpsubscription reports 'Expired' subscriptions the team didn't expect.

Root Cause: Subscriptions exceeded transactional retention (max_distretention) while offline and were marked for reinitialization.

Analysis: Transactional subscriptions expire when their pending commands are cleaned before delivery; Monitor flags them expired.

> **FIX** Reinitialize expired subscriptions and raise retention to accommodate expected Subscriber downtime.

105 Distributor placed on the Publisher (local Distributor) hurts OLTP performance.

Root Cause: A local Distributor competes with the production workload for CPU/IO on the Publisher.

Analysis: Local distribution is simplest but the Log Reader/Distribution overhead lands on the production server.

> **FIX** Move to a remote Distributor to offload replication overhead from the busy Publisher.

106 Distribution database transaction log backup chain conflicts with the SIMPLE recovery expectation.

Root Cause: A monitoring/maintenance plan put the distribution DB into FULL recovery and started log backups it doesn't need.

Analysis: The distribution DB is designed for SIMPLE recovery; forcing FULL adds log-backup overhead and grows the log without benefit.

> **FIX** Return the distribution DB to SIMPLE recovery unless a specific requirement exists, and remove the unnecessary log backup jobs.

Section 7 Latency & Performance

20 scenarios

107 End-to-end latency high; tracer tokens take minutes to reach the Subscriber.

Root Cause: One or more stages (Publisher->Distributor, Distributor->Subscriber) is slow due to volume, blocking, or resource limits.

Analysis: Post a tracer token and read sp_helptracertokens to isolate which leg is slow, then target that stage specifically.

> **FIX** Address the slow leg: tune Log Reader batches, add Subscriber indexes, increase SubscriptionStreams, or resolve blocking.

108 A single very large transaction on the Publisher stalls the whole pipeline.

Root Cause: One massive DML (bulk update/delete) generates millions of commands delivered as one unit.

Analysis: Replication preserves transactional boundaries; a giant transaction is read, stored, and applied atomically, freezing latency until done.

> **FIX** Break large operations into smaller batches, or set -MaxCmdsInTran to chunk delivery of large transactions.

109 Index maintenance on the Publisher floods replication with commands.

Root Cause: REBUILD/REORGANIZE and large UPDATEs on published tables are fully logged and replicated as row changes.

Analysis: Maintenance that touches many rows on published tables becomes replicated commands, spiking distribution volume and latency.

> **FIX** Prefer ONLINE targeted maintenance, schedule off-peak, and consider not publishing tables that need heavy churn, or use partition switching carefully.

110 Subscriber apply is CPU-bound because delivery procs do per-row RPCs.

Root Cause: High command rate with per-row sp_MSins/upd/del RPC overhead saturates Subscriber CPU.

Analysis: Each replicated row change is an RPC on the Subscriber; at tens of thousands/sec this dominates CPU.

> **FIX** Use multiple SubscriptionStreams, ensure good Subscriber indexing, and scale Subscriber hardware; consider batching options.

111 Missing indexes on the Subscriber cause slow UPDATE/DELETE applies.

Root Cause: Delivery procs locate rows by key; without proper indexes each apply scans, crushing throughput.

Analysis: The Subscriber needs indexes supporting the primary-key predicates in sp_MSupd/del. Missing them turns applies into scans.

> **FIX** Create indexes on the Subscriber matching the article keys; verify with actual execution plans of delivery procs.

112 Network bandwidth saturates on WAN links during snapshot delivery.

Root Cause: Uncompressed snapshot BCP files transfer over a constrained WAN, saturating the link.

Analysis: Large initial snapshots move a lot of data; a thin WAN bottlenecks the Distribution Agent and impacts other traffic.

FIX Enable snapshot compression, use init-from-backup, throttle with QoS, or stage snapshots closer to Subscribers.

113 Distribution Agent CommitBatchSize too small, causing excessive round trips.

Root Cause: Default -CommitBatchSize (100) and -CommitBatchThreshold may be suboptimal for the workload/network.

Analysis: Committing in small batches adds transaction overhead and round trips; too large risks long locks. Tuning is workload-specific.

FIX Test higher -CommitBatchSize/-CommitBatchThreshold values to reduce overhead while watching lock duration on the Subscriber.

114 Log Reader -ReadBatchSize too small for a high-volume Publisher.

Root Cause: The Log Reader reads too few transactions per cycle, so it can't keep pace with generation.

Analysis: Publisher-to-Distributor latency climbs when read batches are too small relative to log churn.

FIX Increase -ReadBatchSize and -ReadBatchThreshold and monitor Log Reader throughput vs log growth.

115 Replication agents run continuously and starve other jobs of Agent worker threads.

Root Cause: Many continuous agents consume SQL Server Agent subsystem slots and server resources.

Analysis: SQL Agent has finite subsystem concurrency; too many continuous replication agents crowd out other scheduled work.

FIX Convert some agents to scheduled mode, raise subsystem max worker threads, or distribute agents across servers (pull model).

116 tempdb contention on the Subscriber during large batch applies.

Root Cause: Delivery procs and sorting spill to tempdb; contention on allocation pages slows applies.

Analysis: Heavy apply workloads plus other tempdb users create PFS/GAM/SGAM contention, throttling replication.

FIX Configure multiple equally-sized tempdb files, enable relevant trace flags/defaults, and reduce competing tempdb load.

117 Replication latency correlates with nightly ETL on the Publisher.

Root Cause: Bulk ETL loads into published tables generate replication traffic that collides with the ETL window.

Analysis: ETL that touches published tables doubles as replication load; latency spikes exactly during ETL.

FIX Load into staging (unpublished) tables and switch/merge, or schedule ETL and replication windows to not overlap.

118 Undistributed commands count keeps rising even at low latency readings.

Root Cause: Generation outpaces delivery capacity, so backlog grows despite each command applying quickly.

Analysis: sp_replmonitorsubscriptionpendingcmds shows the true backlog; steady-state throughput must exceed generation to drain it.

FIX Raise delivery parallelism (streams), fix Subscriber bottlenecks, and reduce publisher write volume; add capacity if sustained.

119 Blocking on the Subscriber caused by long-running reporting queries stalls delivery.

Root Cause: Read queries take locks that block replicated writes, or vice versa, since the Subscriber is used for reporting.

Analysis: Reporting on a transactional Subscriber contends with delivery; long selects block sp_MSupd/del applies.

FIX Use READ COMMITTED SNAPSHOT/snapshot isolation on the Subscriber for reporting, or offload reporting to a separate replica.

120 Peer-to-peer replication latency grows as node count increases.

Root Cause: Each P2P node replicates to every other node, so traffic scales with node pairs.

Analysis: P2P is a full mesh; N nodes create N*(N-1) directed streams, multiplying Log Reader/Distribution load.

FIX Limit node count, ensure each node's Distributor is well-resourced, and consider a hub topology if full mesh isn't required.

121 Replication throughput drops after enabling Change Data Capture on the same tables.

Root Cause: CDC and transactional replication share the log reader mechanism, competing for log scan resources.

Analysis: When CDC and replication coexist, the log reader serves both; contention reduces overall throughput.

FIX Right-size the shared log reader, evaluate whether both are needed, and monitor sys.dm_repl* for contention.

122 Distribution database IO latency high, slowing both harvest and delivery.

Root Cause: The distribution DB is on slow storage and is a write-heavy hotspot.

Analysis: Every command is written to and read from distribution; slow disks bottleneck the entire pipeline regardless of tuning elsewhere.

> **FIX** Place the distribution DB (data and log) on fast, low-latency storage and separate data/log volumes.

123 SubscriptionStreams increased but throughput didn't improve.

Root Cause: The bottleneck is elsewhere (Subscriber CPU, disk, blocking) so more streams don't help.

Analysis: Streams help when delivery is the constraint; if the Subscriber or distribution DB is the limit, streams add contention without gain.

> **FIX** Identify the real bottleneck via tracer tokens and wait stats before scaling streams; fix that first.

124 Agent profile changes don't take effect because the agent wasn't restarted.

Root Cause: Distribution/Log Reader profile parameters apply on agent start; a running continuous agent keeps old settings.

Analysis: Editing an agent profile updates stored parameters, but the live agent must restart to read them.

> **FIX** Apply the profile, then stop/start the agent job so the new parameters take effect.

125 Latency reported low in Monitor but users see stale data on the Subscriber.

Root Cause: Monitor latency reflects command delivery, not read consistency; reporting isolation or caching hides fresh data.

Analysis: Delivered != visible if the reader session uses an older snapshot or the app caches. Latency metrics can look fine.

> **FIX** Verify with a tracer token and direct query of a known-updated row; check application caching and isolation levels.

126 Replication agents saturate memory via large in-memory command buffers.

Root Cause: Very large -ReadBatchSize/-CommitBatchSize inflate agent memory, pressuring the server.

Analysis: Oversized batches trade memory for throughput; on constrained servers this causes paging and instability.

> **FIX** Right-size batch parameters to the available memory, and monitor agent process memory during peak load.

Section 8 Schema Changes & DDL

15 scenarios

127 ALTER TABLE ADD COLUMN on the Publisher isn't replicated to the Subscriber.

Root Cause: Schema change replication is disabled (@replicate_ddl=0) or the change type isn't supported for auto-propagation.

Analysis: By default transactional replication propagates many DDL changes, but only when @replicate_ddl is enabled and the change is supported.

> **FIX** Enable sp_changepublication @property='replicate_ddl', @value=1; for unsupported changes, apply manually on both sides or reinitialize.

128 Dropping a replicated column fails: 'Cannot drop column because it is used in replication'.

Root Cause: The column is part of a published article; direct DROP is blocked to protect the article definition.

Analysis: Replication guards published columns. You must remove the column from the article first, or drop via the replication-aware path.

> **FIX** Use sp_dropdisreplicatedcolumn / sp_repldropcolumn (or remove from the article), then drop, propagating the change.

129 Adding a NOT NULL column without default breaks replicated inserts.

Root Cause: The new column exists on the Subscriber but replicated inserts from before the change omit it, or vice versa during rollout.

Analysis: Timing of DDL propagation matters; a window exists where Publisher and Subscriber shapes differ, failing inserts.

> **FIX** Add columns as NULL or with a default, propagate DDL through replication, and coordinate the change window.

130 Renaming a published table breaks replication.

Root Cause: sp_rename on a published object isn't supported and desynchronizes article metadata.

Analysis: Replication keys articles by object; renames aren't propagated and corrupt the mapping.

> **FIX** Drop the article, rename, re-add the article with the new name, and reinitialize; avoid sp_rename on published objects.

131 Adding a new table to an existing publication doesn't reach Subscribers automatically.

Root Cause: New articles require a snapshot to initialize on Subscribers; without regenerating, the new article isn't delivered.

Analysis: Adding an article marks the subscription for a new snapshot of just that article (if immediate_sync/allow_anonymous permit), otherwise reinit is needed.

> **FIX** Add the article, run the Snapshot Agent to generate the new article's snapshot, and let the Distribution Agent apply it.

132 Changing a datatype (e.g. int to bigint) on a replicated column causes delivery failures.

Root Cause: Type change alters command shape and Subscriber compatibility mid-stream.

Analysis: In-flight commands assume the old type; a type change without coordinated propagation breaks applies (overflow/convert errors).

> **FIX** Propagate the ALTER via replicate_ddl, ensure Subscriber column matches, quiesce writes during the change, and validate.

133 Adding an index on the Subscriber is lost after reinitialization.

Root Cause: Custom Subscriber indexes not in the article schema are dropped/recreated by snapshot apply.

Analysis: Reinitialization applies the published schema; anything added manually on the Subscriber outside @schema_option can be wiped.

> **FIX** Include needed indexes via @schema_option (0x... include nonclustered indexes) or use post-snapshot scripts to recreate them.

134 Computed columns replicate their definition but values mismatch.

Root Cause: @schema_option controls whether computed columns replicate as definitions or materialized values; mismatch causes drift.

Analysis: If the Subscriber can't evaluate the computed expression (missing function/column), values diverge or apply fails.

> **FIX** Set @schema_option to replicate computed columns as their evaluated values, or ensure dependencies exist on the Subscriber.

135 TRUNCATE TABLE on a published table is blocked or not replicated.

Root Cause: TRUNCATE is a minimally-logged DDL-like operation and isn't supported/replicated on published tables.

Analysis: Replication needs per-row logging; TRUNCATE bypasses that, so it's disallowed on published tables and won't propagate.

FIX Use DELETE (batched) instead, or unpublish, truncate, and re-add the article with reinitialization.

136 Partition SWITCH on a published table fails.

Root Cause: Partition switching is restricted on replicated tables because it moves data without per-row logging replication can capture.

Analysis: SWITCH is metadata-only; replication can't translate it into row commands, so it's blocked on published tables.

FIX Detach from replication for the operation, or design partitioning outside published tables; reinitialize if the shape changes.

137 Adding a foreign key on the Subscriber not present on the Publisher causes apply failures.

Root Cause: Extra Subscriber constraints reject replicated data that the Publisher allowed.

Analysis: Replication assumes the Subscriber accepts what the Publisher committed; additional constraints can reject valid replicated rows.

FIX Keep Subscriber constraints consistent with the Publisher, or use NOT FOR REPLICATION on triggers/constraints so replication bypasses them.

138 NOT FOR REPLICATION not set on Subscriber triggers, so they fire during apply and cause side effects.

Root Cause: Subscriber triggers execute on replicated DML, duplicating logic or corrupting data.

Analysis: Triggers on the Subscriber fire for replication-applied changes unless created WITH NOT FOR REPLICATION, causing double-processing.

FIX Recreate Subscriber triggers/constraints/identities WITH NOT FOR REPLICATION so replicated operations skip them.

139 sparse columns or column sets on a published table cause snapshot script errors.

Root Cause: Certain column features aren't fully supported by the chosen schema options or Subscriber version.

Analysis: Sparse columns/column sets have replication caveats; scripting them for a down-level or restricted Subscriber fails.

FIX Adjust @schema_option to exclude unsupported clauses, or avoid publishing tables using unsupported column features.

140 DDL change replicated but a dependent view/proc on the Subscriber breaks.

Root Cause: Replication propagates the base table change but not dependent Subscriber-only objects that reference it.

Analysis: Subscriber-side views/procs referencing the changed column become invalid because they aren't part of the article.

> **FIX** Manage Subscriber-only dependent objects separately, updating them alongside replicated DDL via change control.

141 Enabling @replicate_ddl replicates an unintended emergency hotfix DDL to production Subscribers.

Root Cause: Auto DDL propagation forwarded an ad-hoc ALTER made on the Publisher to all Subscribers instantly.

Analysis: With DDL replication on, any supported schema change on the Publisher flows downstream, including mistakes.

> **FIX** Gate Publisher DDL through change control; consider disabling auto DDL and applying schema changes deliberately on both tiers.

Section 9 Identity, Data Integrity & Conflicts

20 scenarios

142 IDENTITY values collide between Publisher and Subscriber in a bidirectional/updatable setup.

Root Cause: Identity range management wasn't configured, so multiple nodes generate overlapping identity values.

Analysis: In multi-writer topologies, uncoordinated identities cause PK collisions. Range management partitions the identity space per node.

> **FIX** Enable automatic identity range management, or manually seed/increment ranges per node so they never overlap.

143 Subscriber identity column reseeds to 1 after reinitialization, causing duplicates.

Root Cause: The snapshot recreated the table with a fresh identity seed instead of preserving the Publisher's current value.

Analysis: If NOT FOR REPLICATION isn't set, the Subscriber may try to assign its own identity values on apply, colliding with replicated ones.

FIX Ensure the identity column uses NOT FOR REPLICATION so replicated inserts carry explicit values, and verify seed alignment.

144 Merge identity range exhaustion despite auto management.

Root Cause: Range size too small for insert velocity, and the threshold to allocate a new range is reached faster than reassignment occurs.

Analysis: High-insert Subscribers burn through ranges; if @identity_range and @pub_identity_range aren't sized right, they hit the ceiling.

FIX Increase @identity_range/@pub_identity_range sizes and lower @threshold so new ranges allocate before exhaustion.

145 tablediff/validation reports row count mismatches between Publisher and Subscriber.

Root Cause: Data divergence from skipped errors, out-of-band Subscriber writes, or missed commands.

Analysis: Row/checksum mismatches confirm the Subscriber is out of sync; the cause is usually prior SkipErrors or manual edits.

FIX Use tablediff to generate a fix script for small drifts, or reinitialize for large divergence; stop the source of drift.

146 sp_publication_validation reports checksum/rowcount failure but data looks identical.

Root Cause: Binary/rowcount validation is sensitive to collation, computed columns, or NULL handling differences.

Analysis: Checksum validation can flag false positives when definitions differ subtly (e.g. computed column materialization) even if user-visible data matches.

FIX Use rowcount-only validation for such tables, or align definitions; investigate specific columns with tablediff.

147 Peer-to-peer conflict: same PK inserted on two nodes simultaneously.

Root Cause: P2P has no built-in conflict resolution by default; concurrent inserts of the same key collide.

Analysis: P2P assumes partitioned writes (each node owns distinct keys). Overlapping writes cause conflicts that halt agents unless detection is enabled.

FIX Partition write ownership by node, enable P2P conflict detection to log/handle conflicts, and design keys to avoid overlap.

148 P2P conflict detection halts the Distribution Agent on a detected conflict.

Root Cause: @conflict_action for P2P is set to stop the agent when a conflict is detected (the safe default).

Analysis: P2P detects conflicts via originator/version metadata; the default action stops delivery to avoid silent divergence.

> **FIX** Resolve the conflicting row manually, choose an appropriate @conflict_action, and restart the agent; fix the write-partitioning root cause.

149 Floating-point/decimal rounding differences flagged as conflicts or validation mismatches.

Root Cause: Different implicit conversions or precision handling between nodes produce tiny value differences.

Analysis: Replication moves exact values, but computed/derived columns evaluated per node can round differently, tripping validation.

> **FIX** Replicate materialized values rather than definitions, standardize precision, and exclude derived columns from strict validation.

150 Merge conflict viewer shows conflicts but resolutions aren't what business expects.

Root Cause: Column-level vs row-level tracking choice and resolver selection don't match the update patterns.

Analysis: Row-level tracking treats any change to a row as conflicting; column-level allows non-overlapping column edits to merge. Wrong choice fabricates conflicts.

> **FIX** Set column-level tracking for tables edited on different columns per node, and pick a resolver matching business rules.

151 Deletes on the Publisher not reflected on the Subscriber (rows linger).

Root Cause: Filtered replication excluded the delete, or the Subscriber row was outside the current filter partition.

Analysis: With row filters, a row that leaves the partition is 'deleted' at the Subscriber via a partition change, but misconfiguration can strand rows.

> **FIX** Verify filter/partition options handle membership changes (@partition_options), and reconcile stranded rows; reinitialize if needed.

152 Subscriber has extra rows the Publisher never sent.

Root Cause: Direct inserts on the Subscriber, or a prior SkipErrors that bypassed deletes, left orphan rows.

Analysis: Transactional Subscribers should be read-only; extra rows indicate out-of-band writes or delivery gaps.

FIX Remove the extra rows (validate first), enforce Subscriber read-only access, and reinitialize to guarantee consistency.

153 NULL vs empty-string or trailing-space differences cause conflict/validation noise.

Root Cause: ANSI_PADDING/collation settings differ, so values that look equal aren't byte-equal.

Analysis: Trailing spaces and padding behavior can differ across servers, producing spurious mismatches in checksum validation.

FIX Standardize ANSI settings and collation, and prefer rowcount validation where padding differences are benign.

154 Merge logical record conflict leaves related rows partially applied.

Root Cause: A conflict within a logical record group resolved inconsistently across member tables.

Analysis: Logical records should apply atomically; a resolver acting per-table can split the group, breaking referential consistency.

FIX Ensure logical records are correctly defined and use a resolver that respects the group, then reconcile affected records.

155 Trigger-based application logic double-applies on the Subscriber, corrupting aggregates.

Root Cause: Subscriber triggers fire on replicated DML because NOT FOR REPLICATION was omitted, re-running business logic.

Analysis: Aggregation/audit triggers that already ran on the Publisher run again on the Subscriber, inflating totals.

FIX Recreate such triggers WITH NOT FOR REPLICATION so they skip replicated operations.

156 Replicated GUID (uniqueidentifier) default NEWID() generates different values per node.

Root Cause: A column defaulting to NEWID() evaluated independently on Publisher and Subscriber diverges if not replicated as a value.

Analysis: Defaults are evaluated where the insert runs; if the article replicates the default rather than the value, GUIDs differ.

FIX Replicate the actual inserted value (not the default definition), or use NEWSEQUENTIALID with care; validate consistency.

157 Datetime precision loss between SQL versions causes off-by-milliseconds mismatches.

Root Cause: datetime vs datetime2 handling or implicit conversion differs across Publisher/Subscriber versions.

Analysis: Rounding of datetime to 3.33ms vs datetime2 precision creates tiny differences flagged by validation.

> **FIX** Use matching datetime types across nodes, or accept/rowcount-validate; standardize the column type in the article.

158 Merge conflict retention purges conflict rows before they can be reviewed.

Root Cause: Conflict tables (MSmerge_conflict_*) are cleaned by retention before operators inspect them.

Analysis: Conflicts are stored temporarily; short retention or cleanup removes them, losing audit detail.

> **FIX** Increase conflict retention or export conflicts promptly; monitor the conflict viewer regularly.

159 Foreign key ordering causes 'conflict with FK constraint' during transactional apply.

Root Cause: Child rows delivered before parents because articles weren't ordered or processed together.

Analysis: Replication generally preserves transaction order, but cross-article timing or parallel streams can violate FK order.

> **FIX** Group related articles, reduce SubscriptionStreams for FK-heavy sets, or use NOT FOR REPLICATION on constraints where safe.

160 Bulk data fixups on the Publisher create a massive conflict/validation backlog in merge.

Root Cause: A large scripted update touched many rows, generating conflicts against Subscriber edits.

Analysis: Mass updates in merge topologies collide with distributed edits, flooding the conflict resolver.

> **FIX** Coordinate bulk changes during a sync freeze, communicate to nodes, and validate/reconcile after; consider reinit for heavy divergence.

161 Checksum validation fails only for LOB (varchar(max)/varbinary(max)) columns.

Root Cause: LOB columns are excluded from or handled differently by binary checksum, or truncated during apply.

Analysis: Some validation methods don't fully checksum LOB data, or a truncation on apply changed the value.

> **FIX** Verify LOB column widths match, use tablediff for LOB-inclusive comparison, and reinitialize the article if truncated.

Section 10 Security & Permissions

13 scenarios

162 Replication agents fail with 'Login failed for user' after a password rotation.

Root Cause: The agent connection credential (SQL login or Windows account) was rotated without updating the agent security settings.

Analysis: Each agent has Publisher/Distributor/Subscriber connection credentials; rotating any without syncing the agent breaks logins.

> **FIX** Update the agent security via sp_changesubscription/sp_change_agent_parameter or the subscription properties with the new credentials.

163 Snapshot Agent can't access the snapshot share: 'Access is denied'.

Root Cause: The Agent process/proxy account lacks share and NTFS permissions on the snapshot folder.

Analysis: Snapshot generation and delivery both touch the share; either the generating account or the delivering account lacks rights.

> **FIX** Grant both the Snapshot Agent and Distribution/Merge Agent accounts read/write on the share and NTFS.

164 Pull Subscriber agent can't connect to the Distributor: 'The Distributor is not available'.

Root Cause: The Subscriber-side agent account lacks a login/access to the distribution database on the Distributor.

Analysis: Pull agents run on the Subscriber and must authenticate to the Distributor's distribution DB to fetch commands.

> **FIX** Grant the pull agent account access to the distribution DB (replication role) and open required network/firewall ports.

165 Agent account not a member of the correct replication role, causing 'permission denied on object'.

Root Cause: The connection account isn't in db_owner or the required replication fixed roles on the relevant database.

Analysis: Agents need specific rights: publication access list membership, distribution DB access, and Subscriber DDL/DML rights.

> **FIX** Add the account to the Publication Access List (sp_grant_publication_access) and grant db_owner where the agent applies data.

166 Publication Access List (PAL) omits an account, blocking a pull Subscriber.

Root Cause: The Subscriber agent login isn't on the PAL, so the Publisher rejects its connection.

Analysis: The PAL controls which logins can access a publication. Missing entries fail pull subscription connections.

> **FIX** Run sp_grant_publication_access to add the login to the PAL for the publication.

167 Windows authentication fails across a domain trust for a remote agent.

Root Cause: Kerberos delegation/SPN issues or a missing trust prevent the Windows agent account from authenticating.

Analysis: Cross-domain replication needs proper trusts and SPNs; double-hop scenarios require delegation.

> **FIX** Register SPNs, configure constrained delegation, verify the trust, or switch the agent to SQL authentication where appropriate.

168 SQL login used by agents is disabled or expired due to a policy.

Root Cause: Password policy/expiration disabled the SQL login the agents use to connect.

Analysis: CHECK_EXPIRATION/CHECK_POLICY on the agent login can disable it, stopping all agents that use it.

> **FIX** Exempt the service login from expiration (or rotate proactively), re-enable it, and update agent security.

169 Distribution DB access via the 'replication' role missing after a Distributor restore.

Root Cause: Restoring the distribution DB didn't restore/remap the logins the agents use.

Analysis: Orphaned users in the restored distribution DB break agent authentication.

> **FIX** Remap logins (ALTER USER ... WITH LOGIN / sp_change_users_login), and reconfigure agent security.

170 Agent uses SQL Server Agent service account but least-privilege changes revoked its rights.

Root Cause: A security hardening removed the Agent account's replication-related permissions.

Analysis: Hardening that strips db_owner/folder rights from the Agent account silently breaks replication agents.

> **FIX** Restore the minimal required rights (snapshot folder, distribution DB, Subscriber DML/DDL) or use dedicated proxy accounts.

171 Proxy account for the replication subsystem is misconfigured, so agent job steps fail.

Root Cause: The SQL Agent proxy mapped to the replication subsystem lacks the underlying credential rights.

Analysis: Agents can run under proxies; if the proxy's credential lacks OS/DB rights, the job step fails at start.

> **FIX** Fix the proxy's credential permissions and ensure the proxy is enabled for the Replication subsystem.

172 TLS/encryption enforced on the Subscriber breaks agent connections.

Root Cause: Force Encryption or certificate trust requirements aren't satisfied by the agent's connection settings.

Analysis: If the endpoint requires encryption and the agent can't validate the cert, the connection fails.

> **FIX** Install/trust the certificate on the agent host, or set the agent connection to trust the server cert appropriately.

173 Firewall blocks the distribution/Subscriber port after a network change.

Root Cause: A new firewall rule or port change blocks the TDS port the agents use.

Analysis: Replication uses standard SQL ports (1433 or custom/dynamic). Blocked ports cause connection timeouts, not auth errors.

> **FIX** Open the required static ports on the Distributor/Subscriber, and prefer static ports for replicated instances.

174 Sensitive data replicated to a lower-security Subscriber violates compliance.

Root Cause: Column filtering/masking wasn't applied, so regulated columns (PII/PCI) flow to a less-controlled environment.

Analysis: Replication moves data as-is; without vertical filters or masking, sensitive columns leave the compliant boundary.

> **FIX** Use column (vertical) filters to exclude sensitive columns, or apply masking/transformation before replication; document the scope for audit.

Section 11 Reinitialization & Recovery

12 scenarios

175 Subscription marked for reinitialization but no new snapshot is generated.

Root Cause: The subscription was flagged (sp_reinitsubscription) but the Snapshot Agent wasn't run to produce a fresh snapshot.

Analysis: Reinitialization only queues the request; a snapshot must exist and be applied for the Subscriber to resync.

> **FIX** Run the Snapshot Agent, then start the Distribution/Merge Agent to apply the new snapshot to the flagged subscription.

176 Reinitializing a huge publication causes hours of downtime and blocking.

Root Cause: Full snapshot regeneration and reapply of large tables locks and transfers everything again.

Analysis: Reinit is expensive for big publications; it re-scripts schema, re-BCPs data, and reapplies constraints/indexes.

> **FIX** Prefer targeted article reinit, init-from-backup, or tablediff repair for small drift instead of a full publication reinit.

177 After Publisher failover in an FCI, agents can't find the Publisher.

Root Cause: The virtual network name is correct but agents/metadata reference a node name, or a brief outage stalled agents.

Analysis: FCI presents a single VNN; if replication was configured against a node name, failover breaks connectivity.

> **FIX** Configure replication against the FCI virtual network name and ensure agents retry; validate connectivity post-failover.

178 Restoring the Publisher database loses replication configuration.

Root Cause: The restore didn't use KEEP_REPLICATION, so replication metadata in the database was discarded.

Analysis: A normal restore removes replication settings from the user DB; KEEP_REPLICATION preserves them for supported scenarios.

> **FIX** Restore WITH KEEP_REPLICATION where applicable, or reconfigure replication and reinitialize after the restore.

179 Point-in-time restore of the Publisher creates command divergence with the Distributor.

Root Cause: Restoring to an earlier LSN than the Distributor already received re-sends or skips commands.

Analysis: The Distributor holds commands beyond the restored point; the log chain no longer matches, causing duplicates or gaps.

> **FIX** Reinitialize subscriptions after a point-in-time restore to guarantee consistency; don't rely on the old command stream.

180 Distributor lost; need to rebuild without full reinitialization.

Root Cause: The distribution database was lost and there's no backup, so the command pipeline is gone.

Analysis: Without the distribution DB, in-flight commands are unrecoverable; the safest path is rebuild plus reinitialize.

> **FIX** Recreate the Distributor and distribution DB, reconfigure publications/subscriptions, and reinitialize from a new snapshot.

181 sp_repldone used incorrectly skips undelivered transactions, losing data at Subscribers.

Root Cause: An operator ran sp_repldone to free the log without ensuring commands were delivered, discarding pending changes.

Analysis: sp_repldone marks transactions as distributed; misuse tells replication to forget commands that never reached Subscribers.

> **FIX** Only use sp_repldone as a last resort with full understanding; afterward validate and reinitialize to restore consistency.

182 Reinitialization loop: subscription keeps re-requesting a snapshot every sync.

Root Cause: A recurring apply failure marks the subscription failed, and automation re-flags it for reinit each cycle.

Analysis: If the underlying error (permission, schema, constraint) isn't fixed, each reinit fails and re-triggers, looping.

> **FIX** Diagnose and fix the root apply error first, then reinitialize once; disable auto-reinit until resolved.

183 Detaching/attaching a published database corrupts replication metadata.

Root Cause: Detach/attach isn't a supported way to move a published database and can break replication linkage.

Analysis: Replication metadata and log chain assumptions don't survive detach/attach cleanly.

> **FIX** Use backup/restore WITH KEEP_REPLICATION for supported moves, or remove and reconfigure replication after the move.

184 Upgrading SQL Server leaves replication in a broken state post-upgrade.

Root Cause: Replication wasn't quiesced before the upgrade, or agents/version rules weren't respected across tiers.

Analysis: Upgrades should follow Distributor-first ordering and drain pending commands; skipping this desynchronizes tiers.

FIX Follow Microsoft's upgrade order (Distributor, then Publisher, then Subscriber), drain commands first, and validate afterward.

185 Reinitialize with 'no snapshot' option leaves Subscriber schema/data unchanged but resets metadata.

Root Cause: @invalidate_snapshot=0 / 'replication support only' was used, assuming data already matches, but it didn't.

Analysis: The no-snapshot reinit trusts that Publisher and Subscriber already match; if they don't, silent divergence persists.

FIX Only use no-snapshot reinit when you've verified data equivalence (e.g. via a restored backup); otherwise use a full snapshot.

186 Disaster recovery failover to a DR Subscriber then promoted, but replication can't be reversed easily.

Root Cause: The topology wasn't designed for role reversal, so making the old Subscriber a Publisher requires rebuild.

Analysis: Standard transactional replication is one-directional; DR promotion needs pre-planned bidirectional or P2P design.

FIX Design DR with reversible topologies (P2P or documented rebuild runbooks); after promotion, reconfigure and reinitialize as planned.

Section 12 Peer-to-Peer, AG Coexistence & Misc

14 scenarios

187 Peer-to-peer setup requires identical schema on all nodes but a node drifted.

Root Cause: P2P mandates matching table structures across nodes; a manual change on one node broke the requirement.

Analysis: P2P replicates in all directions and assumes structural parity; drift causes apply failures and conflicts.

FIX Realign schema across all P2P nodes (use replicated DDL), quiesce, and validate before resuming.

188 Adding a node to an existing P2P topology fails without quiescing.

Root Cause: P2P topology changes historically require quiescing (draining) the system unless using the online add-node feature correctly.

Analysis: Enterprise supports adding nodes without full quiesce in newer versions, but prerequisites (originator ids, backups) must be met.

> **FIX** Follow the P2P add-node procedure: back up, ensure @p2p_continue_onconflict settings, and use the wizard's online option where supported.

189 P2P originator_id collision after cloning a node from another.

Root Cause: Cloning copied the originator id, so two nodes claim the same identity, confusing conflict detection.

Analysis: Each P2P node needs a unique originator id; duplicates make the system unable to distinguish sources.

> **FIX** Assign unique originator ids per node when configuring P2P; never clone a configured node without re-identifying it.

190 Replication Publisher in an Availability Group fails to run agents after failover.

Root Cause: Publisher redirection to the AG listener wasn't configured, so agents still target the failed primary.

Analysis: Replication is AG-aware only with sp_redirect_publisher mapping the original Publisher to the listener; otherwise failover breaks it.

> **FIX** Configure the AG listener as the redirected Publisher (sp_redirect_publisher) and validate with sp_validate_replica_hosts_as_publishers.

191 Distribution database can't be placed in an Availability Group in older versions.

Root Cause: Support for the distribution database in an AG is version-dependent; older builds don't allow it.

Analysis: Making the Distributor HA required an FCI historically; AG support for the distribution DB came later.

> **FIX** Use a supported version for distribution-in-AG, or make the Distributor highly available via FCI/clustered storage.

192 Subscriber is an AG secondary and becomes read-unavailable during failover, stalling delivery.

Root Cause: The Subscriber AG failed over and the listener/read-write mapping for the agent wasn't updated.

Analysis: If the Subscriber is in an AG, agents must target the listener; a failover without proper routing stalls applies.

> **FIX** Point the subscription at the Subscriber AG listener and ensure the target is writable post-failover.

193 Log Reader can't keep up because the published DB is also a busy AG primary.

Root Cause: AG synchronization and the Log Reader both read the log heavily, competing for resources.

Analysis: An AG primary that's also a replication Publisher has double log-read pressure, raising Publisher-to-Distributor latency.

> **FIX** Ensure ample IO/CPU headroom, tune Log Reader batches, and consider a remote Distributor to offload processing.

194 Mixed replication + mirroring legacy setup breaks on mirroring failover.

Root Cause: Database mirroring failover isn't transparently handled by replication without the documented mirroring-Publisher configuration.

Analysis: Mirroring (deprecated) required specific replication settings to survive failover; missing them breaks agents.

> **FIX** Migrate to Availability Groups with publisher redirection, or apply the documented mirroring+replication configuration if still on mirroring.

195 Replication Monitor shows agents as 'Retrying' repeatedly without a clear error.

Root Cause: Transient connectivity or a recurring low-severity failure keeps agents in retry loops.

Analysis: Agents auto-retry on many errors; a flapping network or intermittent lock causes visible but non-fatal retrying.

> **FIX** Inspect agent output files (-Output) for the real error, stabilize the network/locking, and tune retry counts.

196 Agent output/log files fill the disk over time.

Root Cause: Agents configured with verbose -Output logging to files that are never rotated or purged.

Analysis: Persistent -Output at high verbosity accumulates large logs, eventually filling the volume.

> **FIX** Lower verbosity, rotate/purge agent output files, and only enable verbose logging temporarily for diagnosis.

197 Replication interferes with a planned database migration to Azure SQL Database.

Root Cause: Azure SQL Database supports being a push Subscriber (with caveats) but not a Publisher/Distributor; topology assumptions break.

Analysis: PaaS targets have replication limitations; treating Azure SQL DB like a full instance for replication fails.

FIX Use Azure SQL DB only in supported replication roles (e.g. Subscriber), or use Data Sync/Managed Instance for broader needs.

198 Heterogeneous replication to Oracle/other RDBMS deprecated and now failing.

Root Cause: Non-SQL Server Publishers/Subscribers via replication are deprecated/removed in modern versions.

Analysis: Legacy heterogeneous replication (e.g. Oracle Publishing) is no longer supported, so upgrades break these flows.

FIX Migrate heterogeneous flows to SSIS, ADF, or CDC-based pipelines; don't rely on deprecated heterogeneous replication.

199 Replication agents fail after a server rename (@@SERVERNAME mismatch).

Root Cause: The Windows/SQL server was renamed but sysservers/@@SERVERNAME wasn't updated, so replication references the old name.

Analysis: Replication hardcodes server names in metadata; a rename without sp_dropserver/sp_addserver breaks all agent connections.

FIX Correct @@SERVERNAME via sp_dropserver/sp_addserver and restart SQL, then fix replication metadata or reconfigure.

200 Time zone / clock skew between nodes affects merge conflict resolution and tokens.

Root Cause: Unsynchronized system clocks skew timestamp-based conflict resolution and tracer token timing.

Analysis: Merge resolvers and latency measurements can rely on time; skew produces wrong winners or misleading latency.

FIX Synchronize clocks via NTP across all nodes and avoid purely time-based resolvers where skew is possible.

End of Reference — 200 Scenarios | **Praveen Madupu**